MILESTONE 1

Project Proposal & Data Preparation

CoffeeKing, Nashville Yelp Academic Dataset

Client	CoffeeKing (fictitious startup coffee company)
Dataset	Yelp Academic Dataset, Filtered for Nashville, TN
Course	SQL for Data Science Capstone, UC Davis / Coursera
Milestone	1: Project Proposal and Data Selection / Preparation
Tool	Databricks Free Edition, SQL Notebook

Step 1: Preparing for Your Proposal

1.1 Client & Dataset Selection

I selected CoffeeKing as my client and the Yelp Academic Dataset (filtered for Nashville, TN) as the primary data source. CoffeeKing is a new startup coffee company aiming to provide a unique and novel experience to its customers in Nashville, Tennessee, wanting to appeal to a wide variety of clientele.

This client was chosen for three key reasons:

- Richness of the data: The Yelp dataset provides detailed business attributes, customer reviews, tips, and check-in records, all directly relevant to strategic decisions about coffee shop placement and operations.
- Clear business questions: Location selection, hours of operation, and customer sentiment are natural, answerable questions given the available columns (stars, review_count, hours, postal_code, categories).
- SQL suitability: The relational structure (business, review, tip, checkin, user tables linked by business_id and user_id) is ideal for JOIN-based SQL analysis.

1.2 Data Import and Cleaning Steps

The dataset was provided as a ZIP archive containing five CSV files. The following steps were taken to import and prepare the data in Databricks Free Edition:

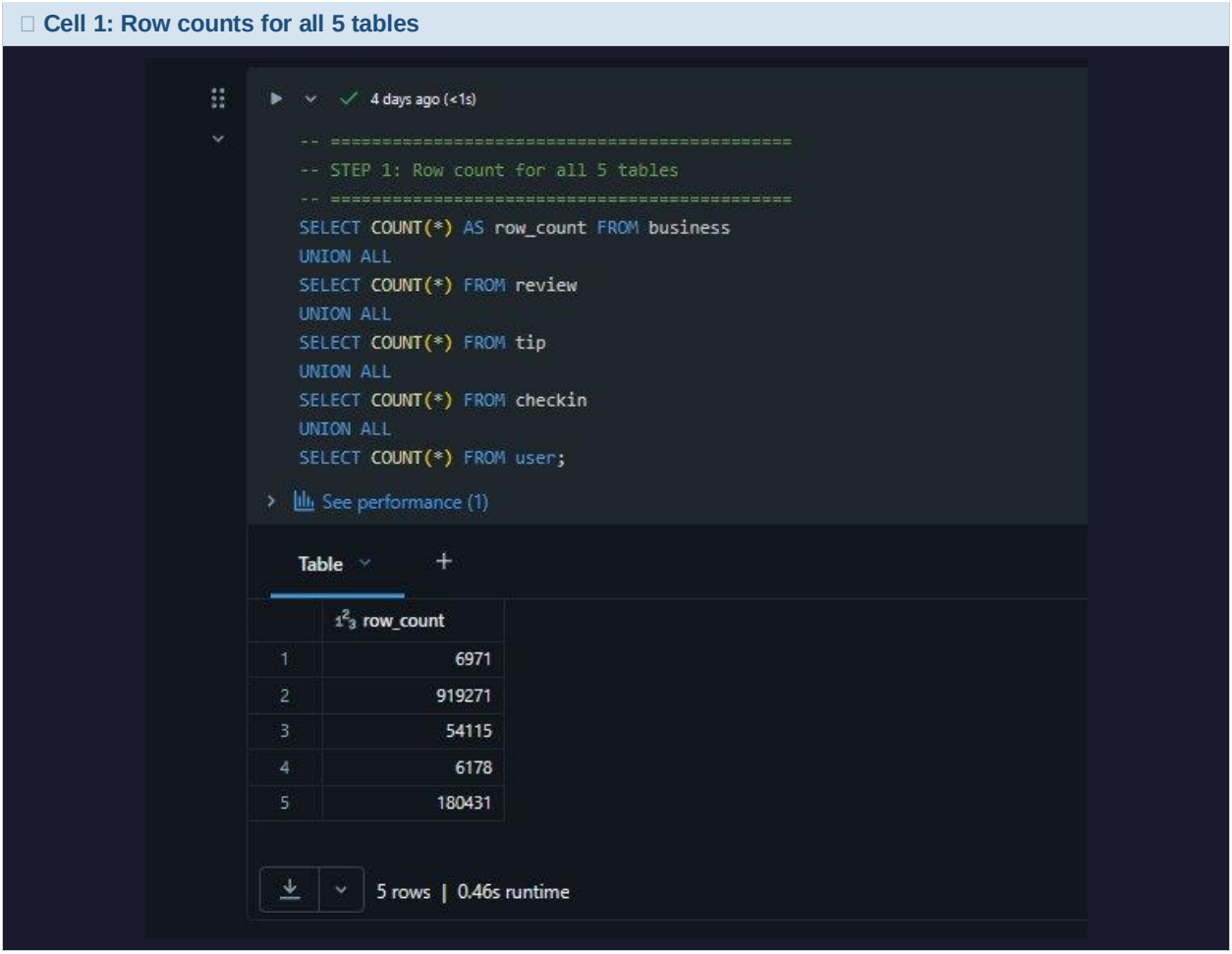
1. Extracted the ZIP archive and uploaded all five CSV files into Databricks using the File Upload interface. Databricks auto-created Delta tables for each CSV.
2. Identified and excluded the _rescued_data column a Databricks auto-generated artifact containing only NULL/empty string values with no analytical value.
3. Profiled each table using DESCRIBE TABLE and COUNT queries to confirm row counts, column names, and data types before any analysis.
4. Noted that postal_code was stored as bigint (numeric) instead of string. Applied CAST(postal_code AS STRING) in all location queries to preserve zip code formatting.

5. Verified the hours and attributes columns contain JSON-formatted strings embedded within CSV cells; handled via LIKE pattern matching rather than full JSON parsing.
6. Identified 285 rows in the business table with missing address values (has_address = 6,686 out of 6,971). These are excluded from address-dependent queries.
7. Confirmed star ratings contain only valid Yelp values (1.0 to 5.0 in 0.5 increments) zero data integrity issues found in that column.

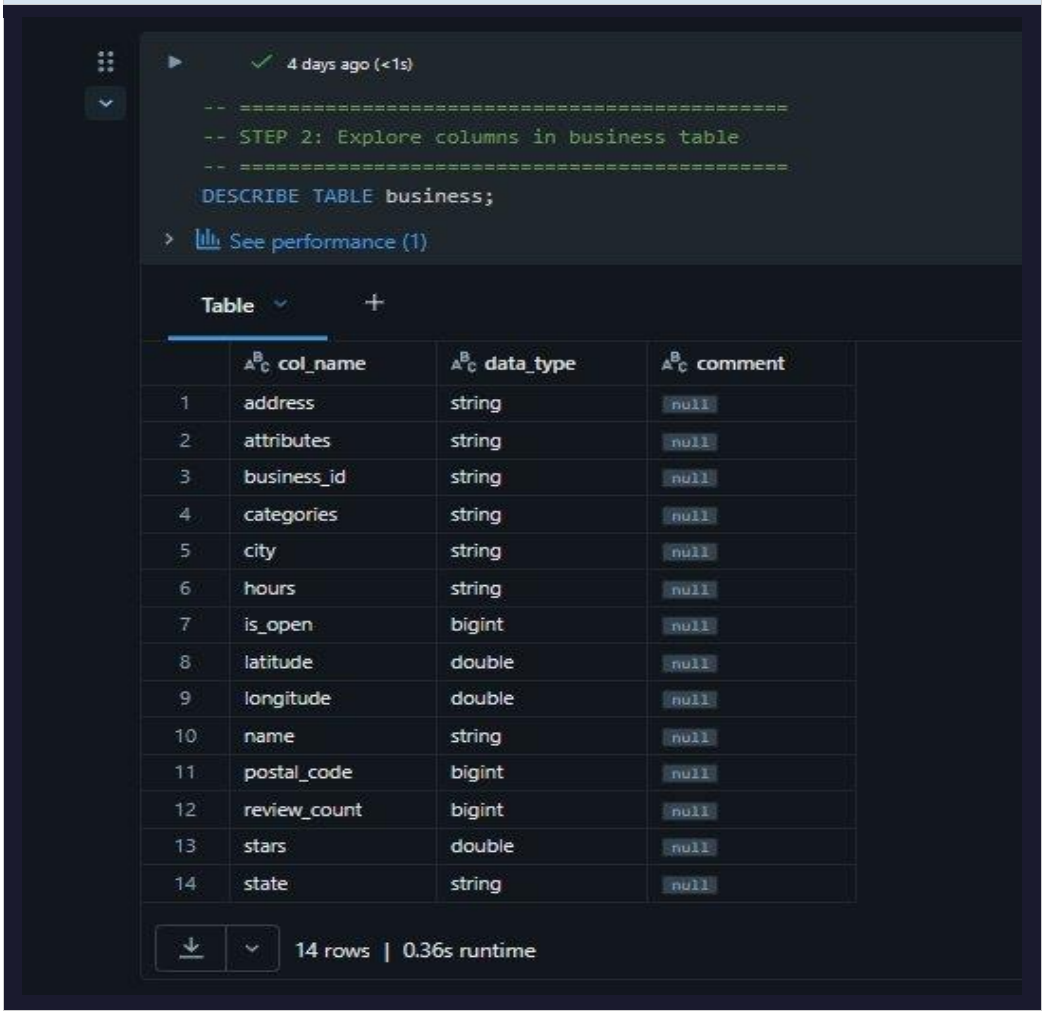
1.3 Initial Data Exploration

The queries below were executed in a Databricks SQL Notebook. Each cell shows the query and its result together, providing a reproducible record of the exploration process.

Cell 1: Row counts for all 5 tables



Cell 2: Column names and data types (DESCRIBE TABLE)



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```
-- ============================================
-- STEP 2: Explore columns in business table
-- ============================================
DESCRIBE TABLE business;
```

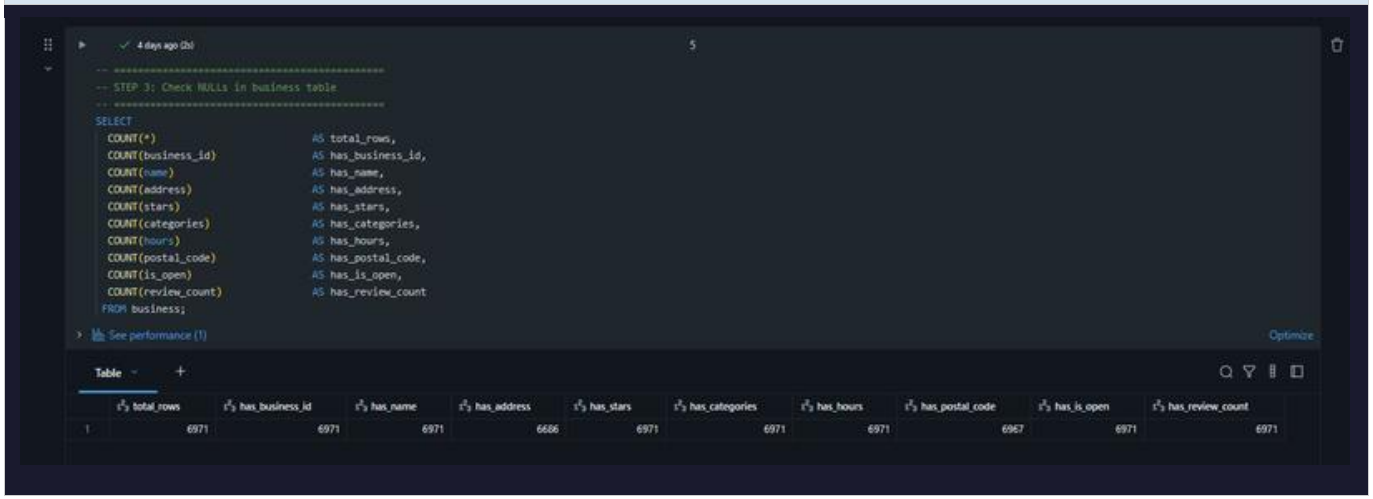
See performance (1)

Table +

	col_name	data_type	comment
1	address	string	null
2	attributes	string	null
3	business_id	string	null
4	categories	string	null
5	city	string	null
6	hours	string	null
7	is_open	bigint	null
8	latitude	double	null
9	longitude	double	null
10	name	string	null
11	postal_code	bigint	null
12	review_count	bigint	null
13	stars	double	null
14	state	string	null

14 rows | 0.36s runtime

Cell 3: NULL value check across all key columns



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```
-- ============================================
-- STEP 3: Check NULLs in business table
-- ============================================
SELECT
  COUNT(*)                AS total_rows,
  COUNT(business_id)      AS has_business_id,
  COUNT(name)             AS has_name,
  COUNT(address)          AS has_address,
  COUNT(stars)            AS has_stars,
  COUNT(categories)       AS has_categories,
  COUNT(hours)            AS has_hours,
  COUNT(postal_code)      AS has_postal_code,
  COUNT(is_open)          AS has_is_open,
  COUNT(review_count)     AS has_review_count
FROM business;
```

See performance (1) Optimize

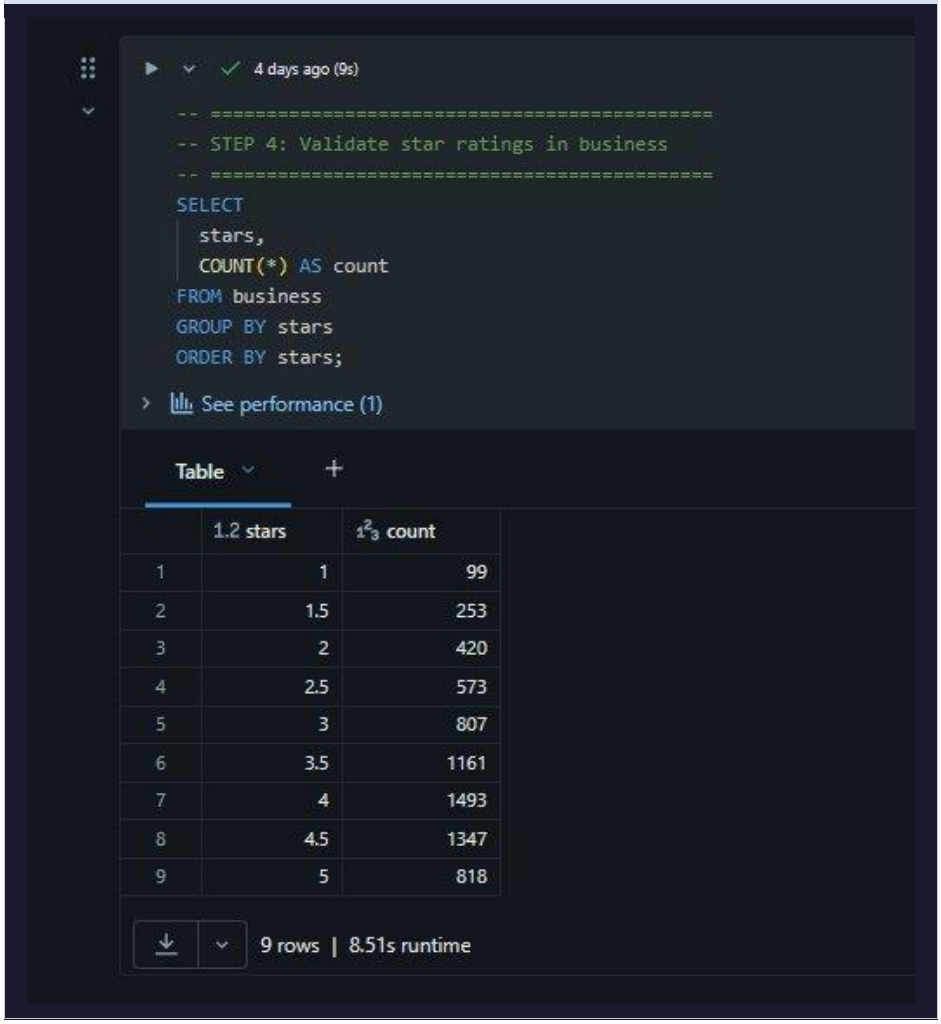
Table +

	total_rows	has_business_id	has_name	has_address	has_stars	has_categories	has_hours	has_postal_code	has_is_open	has_review_count
1	6971	6971	6971	6686	6971	6971	6971	6967	6971	6971

Key findings from the NULL check:

- address: 285 rows missing (6,686 of 6,971) excluded from address specific queries
- postal_code: 4 rows missing (6,967 of 6,971) negligible impact
- All other key columns (stars, categories, hours, is_open, review_count): 100% complete

Cell 4: Star rating validation confirms only valid Yelp values exist



Star rating distribution across all 6,971 Nashville businesses shows a positive skew 4.0 stars is the most common rating (1,493 businesses), confirming data validity and healthy customer review behaviour.

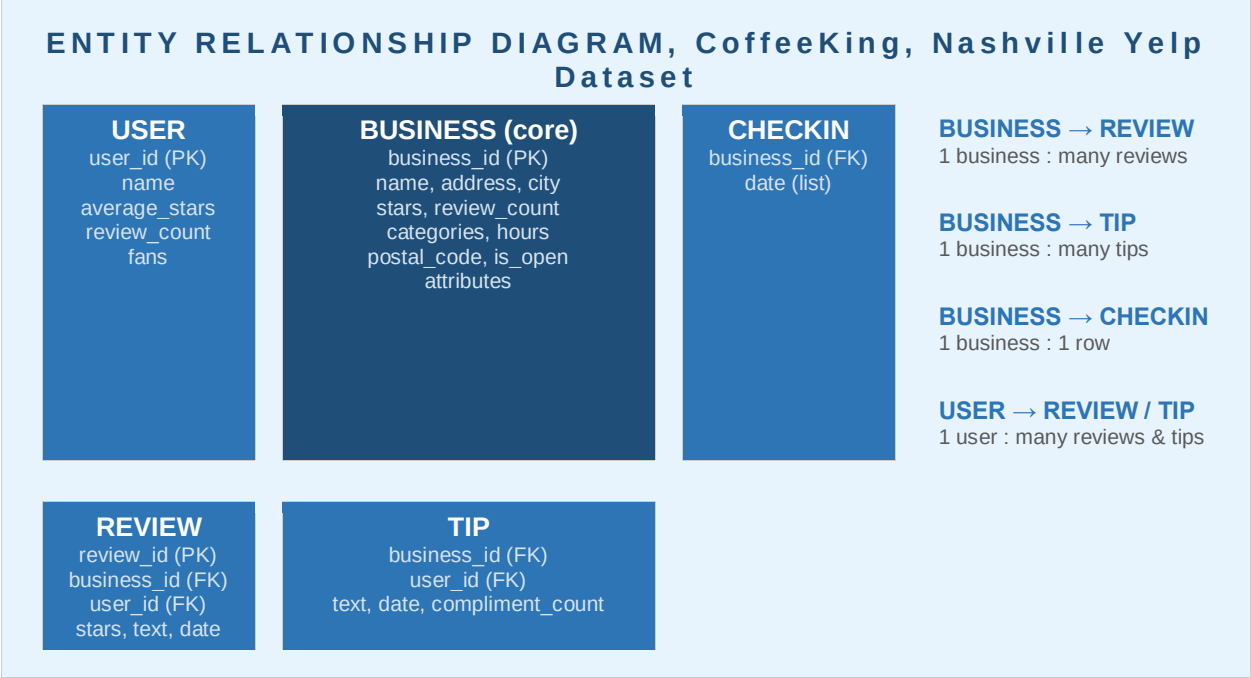
Summary statistics for the full dataset and coffee subset:

Table	Row Count	Key Columns	Linked By
business	6,971	business_id, name, stars, categories, hours, postal_code	business_id
review	919,271	review_id, business_id, user_id, stars, text, date	business_id + user_id

Table	Row Count	Key Columns	Linked By
tip	54,115	business_id, user_id, text, date, compliment_count	business_id + user_id
checkin	6,178	business_id, date (multi-value list)	business_id
user	180,431	user_id, name, average_stars, review_count, fans	user_id

1.4 Entity Relationship Diagram (ERD)

All five tables connect through the central BUSINESS table via business_id. The USER table additionally connects to both REVIEW and TIP via user_id. The CHECKIN table links solely to BUSINESS.



Step 2: Project Proposal

2.1 Project Description

This project leverages the Yelp Academic Dataset (Nashville, TN) to provide data-driven insights for CoffeeKing, a new coffee startup seeking to establish a competitive presence in the Nashville market. The dataset contains rich information about business locations, customer ratings, review volumes, operating hours, and customer behaviour across 6,971 businesses, of which 394 fall into the coffee and cafe category.

Analysis was conducted using Databricks Free Edition, SQL Notebooks, applying structured SQL queries across five relational tables: business, review, tip, checkin, and user. This approach delivers actionable recommendations on where CoffeeKing should open, what hours to operate, and how to position itself in the Nashville coffee market.

The findings are relevant to CoffeeKing's management team, investors, and operations staff. Secondary audiences include urban planners, franchise consultants, and retail analysts interested in how consumer review data can inform hospitality industry strategy.

2.2 Research Questions

The following three questions guided the analysis:

8. **Q1.** Which Nashville zip codes have the highest average star ratings for coffee shops, and how does review volume and closure rate vary by location? (Location selection strategy)
9. **Q2.** Is there a relationship between a coffee shop's review volume and its average star rating? Do high-review-count businesses score higher or regress toward the mean?
10. **Q3.** What operating schedule do top-rated Nashville coffee shops share? Do shops open more days per week earn higher or lower customer ratings?

2.3 Hypotheses

The following assumptions were proposed before analysis and tested against the data:

11. **H1.** Downtown Nashville zip codes (e.g., 37203) likely have more coffee shops due to foot traffic, but increased competition and tourist volume may result in lower average ratings compared to residential neighbourhoods.
12. **H2.** Coffee shops with very high review counts (500+) will tend to regress toward the mean, clustering around 3.5 to 4.0 stars, while lesser-known shops may show more extreme ratings.
13. **H3.** Coffee shops open 7 days a week will earn higher ratings than those with restricted schedules, as greater availability correlates with customer satisfaction.

2.4 Analytical Approach & Results

Analysis was performed in 11 SQL cells within a Databricks notebook. Each cell is documented below with its query result screenshot.

Cell 5: Coffee subset overview: 394 shops, avg 3.7 stars, 128 closed (33%)

```
-- ============================================
-- STEP 5: Count coffee/cafe businesses
-- ============================================
SELECT
  COUNT(*) AS total_coffee_businesses,
  ROUND(AVG(stars), 2) AS avg_stars,
  MIN(stars) AS min_stars,
  MAX(stars) AS max_stars,
  SUM(review_count) AS total_reviews,
  SUM(CASE WHEN is_open = 1 THEN 1 ELSE 0 END) AS currently_open,
  SUM(CASE WHEN is_open = 0 THEN 1 ELSE 0 END) AS permanently_closed
FROM business
WHERE LOWER(categories) LIKE '%coffee%'
  OR LOWER(categories) LIKE '%cafe%'
  OR LOWER(categories) LIKE '%cafes%'
  OR LOWER(categories) LIKE '%coffee roasters%';
```

4 days ago (1s) 8

See performance (1)

Table +

	total_coffee_businesses	avg_stars	min_stars	max_stars	total_reviews	currently_open	permanently_closed
1	394	3.7	1	5	37881	266	128

1 row | 0.91s runtime

Cell 6: Top 10 most-reviewed coffee shops competitor landscape

4 days ago (1s) 10

```sql
-- ============================================
-- STEP 6: Top 10 coffee shops by review count
-- ============================================
SELECT
  name,
  stars,
  review_count,
  postal_code,
  is_open,
  CASE WHEN is_open = 1 THEN 'Open' ELSE 'Closed' END AS status
FROM business
WHERE LOWER(categories) LIKE '%coffee%'
   OR LOWER(categories) LIKE '%cafe%'
   OR LOWER(categories) LIKE '%cafes%'
   OR LOWER(categories) LIKE '%coffee roasters%'
ORDER BY review_count DESC
LIMIT 10;
```

See performance (1)

Table

	name	1.2 stars	review_count	postal_code	is_open	status
1	Milk and Honey Nashvi...	4	1725	37203	1	Open
2	Fido	4	1260	37212	1	Open
3	Crema Coffee Roasters	4.5	998	37210	1	Open
4	Frothy Monkey	4	988	37219	1	Open
5	Frothy Monkey	4	852	37204	1	Open
6	Sky Blue Cafe	4	836	37206	1	Open
7	Another Broken Egg Cafe	4	818	37201	1	Open
8	The Garden Brunch Cafe	4.5	712	37208	0	Closed
9	Germantown Cafe	4	662	37208	1	Open
10	Barista Parlor	3.5	600	37206	1	Open

10 rows | 0.94s runtime

Cell 7: Star rating distribution for coffee shops (window function)

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```sql
-- ============================================
-- STEP 7: Star rating distribution (coffee only)
-- ============================================
SELECT
  stars,
  COUNT(*) AS shop_count,
  ROUND(COUNT(*) * 100.0 / SUM(COUNT(*)) OVER(), 1) AS percentage
FROM business
WHERE LOWER(categories) LIKE '%coffee%'
   OR LOWER(categories) LIKE '%cafe%'
   OR LOWER(categories) LIKE '%cafes%'
   OR LOWER(categories) LIKE '%coffee roasters%'
GROUP BY stars
ORDER BY stars;
```

See performance (1)

Table Visualization 1

	1.2 stars	shop_count	.00 percentage
1	1	3	0.8
2	1.5	14	3.6
3	2	26	6.6
4	2.5	23	5.8
5	3	34	8.6
6	3.5	58	14.7
7	4	110	27.9
8	4.5	100	25.4
9	5	26	6.6

9 rows | 14.94s runtime

53.3% of Nashville coffee shops are rated 4.0 or above. The distribution leans strongly positive, with 4.0 stars being the most common rating (27.9% of shops). This confirms the market rewards quality with strong ratings.

Research Question 1: Best zip codes for CoffeeKing:

Cell 8: Q1: Avg stars and shop counts by zip code (HAVING COUNT >= 3)

```
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-- ============================================
-- STEP 8: Q1 - Best zip codes for coffee shops
-- ============================================
SELECT
  CAST(postal_code AS STRING)        AS zip_code,
  COUNT(*)                           AS total_shops,
  ROUND(AVG(stars), 2)               AS avg_stars,
  SUM(review_count)                  AS total_reviews,
  SUM(CASE WHEN is_open = 1
      THEN 1 ELSE 0 END)             AS open_shops,
  SUM(CASE WHEN is_open = 0
      THEN 1 ELSE 0 END)             AS closed_shops
FROM business
WHERE LOWER(categories) LIKE '%coffee%'
   OR LOWER(categories) LIKE '%cafe%'
   OR LOWER(categories) LIKE '%cafes%'
   OR LOWER(categories) LIKE '%coffee roasters%'
GROUP BY postal_code
HAVING COUNT(*) >= 3
ORDER BY avg_stars DESC, total_reviews DESC;

See performance (1)
```

Table +

	zip_code	total_shops	avg_stars	total_reviews	open_shops	closed_shops
1	37220	3	4.67	93	2	1
2	37206	31	4.24	4125	19	12
3	37216	13	4.19	626	9	4
4	37208	19	4.16	3102	10	9
5	37204	22	3.77	3502	16	6
6	37201	15	3.77	3052	10	5
7	37209	28	3.77	2250	22	6
8	37219	21	3.76	1961	12	9
9	37210	11	3.68	1869	8	3
10	37212	21	3.67	3200	12	9
11	37203	78	3.65	7034	56	22
12	37215	15	3.63	1212	12	3
13	37211	30	3.53	1663	17	13
14	37221	12	3.46	596	8	4
15	37205	12	3.46	337	8	4
16	37214	30	3.28	2037	21	9
17	37207	8	3.25	338	7	1
18	37217	8	3	343	7	1
19	37228	3	2	113	3	0

Result: Zip code 37203 (downtown) has the most shops (78) but one of the lowest ratings (3.65), confirming Hypothesis 1. Zip codes 37206 (4.24 stars, 31 shops) and 37216 (4.19 stars, lowest closure rate at 30.8%) are the strongest candidates for CoffeeKing's first location.

Research Question 2: Review volume vs star rating:

Cell 9: Q2: CASE WHEN bucketing of review count vs average stars

```
02:50 PM (16s)
-- ============================================
-- STEP 9: Q2 - Review volume vs star rating
-- ============================================
SELECT
  CASE
    WHEN review_count < 50   THEN '1. Low (under 50)'
    WHEN review_count < 200  THEN '2. Mid (50-199)'
    WHEN review_count < 500  THEN '3. High (200-499)'
    ELSE                          '4. Very High (500+)'
  END                         AS review_bucket,
  COUNT(*)                    AS shop_count,
  ROUND(AVG(stars), 2)        AS avg_stars,
  MIN(stars)                  AS min_stars,
  MAX(stars)                  AS max_stars,
  ROUND(AVG(review_count), 0) AS avg_reviews_in_bucket
FROM business
WHERE LOWER(categories) LIKE '%coffee%'
   OR LOWER(categories) LIKE '%cafe%'
   OR LOWER(categories) LIKE '%cafes%'
   OR LOWER(categories) LIKE '%coffee roasters%'
GROUP BY review_bucket
ORDER BY review_bucket;
```

See performance (1)

Table +

	review_bucket	shop_count	avg_stars	min_stars	max_stars	avg_reviews_in_bucket
1	1. Low (under 50)	231	3.53	1	5	23
2	2. Mid (50-199)	116	3.89	1	5	98
3	3. High (200-499)	34	4.04	3.5	5	299
4	4. Very High (500+)	13	4.12	3.5	4.5	859

Result: Hypothesis 2 was disproved. Rather than regressing to the mean, shops with 500+ reviews score the highest (4.12 stars), while low-review shops score lowest (3.53). Quality and popularity rise together good shops earn both more reviews and better ratings.

Research Question 3: Operating schedule vs star rating:

Cell 10: Q3: Operating schedule type vs avg star rating

02:55 PM (2s)

```
-- ============================================
-- STEP 10: Q3 - Opening hours vs star rating
-- ============================================
SELECT
  CASE
    WHEN hours LIKE '%"Monday":"%'
     AND hours LIKE '%"Saturday":"%'
     AND hours LIKE '%"Sunday":"%'   THEN '7 days a week'
    WHEN hours LIKE '%"Monday":"%'
     AND hours LIKE '%"Saturday":"%' THEN 'Mon-Sat (6 days)'
    WHEN hours LIKE '%"Monday":"%'   THEN 'Weekdays only'
    ELSE 'Other/Irregular'
  END                              AS schedule_type,
  COUNT(*)                         AS shop_count,
  ROUND(AVG(stars), 2)             AS avg_stars,
  SUM(review_count)                AS total_reviews,
  SUM(CASE WHEN is_open = 1
       THEN 1 ELSE 0 END)          AS currently_open
FROM business
WHERE LOWER(categories) LIKE '%coffee%'
   OR LOWER(categories) LIKE '%cafe%'
   OR LOWER(categories) LIKE '%cafes%'
   OR LOWER(categories) LIKE '%coffee roasters%'
GROUP BY schedule_type
ORDER BY avg_stars DESC;
```

See performance (1)

Table +

	schedule_type	shop_count	avg_stars	total_reviews	currently_open
1	Mon-Sat (6 days)	37	4.26	3305	21
2	Weekdays only	22	4.05	1169	10
3	Other/Irregular	56	3.84	3418	24
4	7 days a week	279	3.57	29989	211

Result: Hypothesis 3 was disproved. 7-day-a-week shops score the lowest (3.57 stars) while Mon-Sat shops score highest (4.26 stars). Smaller, focused independent shops operating 6 days outperform high-volume 7-day chains on quality perception.

Final multi-criteria recommendation for CoffeeKing:

Cell 11: Final summary: top zip codes ranked by avg stars and closure rate

```sql
-- ============================================================
-- STEP 11: Final summary - Top target zip codes for CoffeeKing with full context
-- ============================================================
SELECT
  CAST(postal_code AS STRING)          AS zip_code,
  COUNT(*)                             AS total_shops,
  ROUND(AVG(stars), 2)                 AS avg_stars,
  SUM(review_count)                    AS total_reviews,
  ROUND(AVG(review_count), 0)          AS avg_reviews_per_shop,
  SUM(CASE WHEN is_open = 1
      THEN 1 ELSE 0 END)               AS open_shops,
  SUM(CASE WHEN is_open = 0
      THEN 1 ELSE 0 END)               AS closed_shops,
  ROUND(SUM(CASE WHEN is_open = 0
      THEN 1 ELSE 0 END) * 100.0 /
      COUNT(*), 1)                     AS closure_rate_pct
FROM business
WHERE LOWER(categories) LIKE '%coffee%'
   OR LOWER(categories) LIKE '%cafe%'
   OR LOWER(categories) LIKE '%cafes%'
   OR LOWER(categories) LIKE '%coffee roasters%'
GROUP BY postal_code
HAVING COUNT(*) >= 3
ORDER BY avg_stars DESC, closure_rate_pct ASC
LIMIT 8;
```

See performance (1)

Table

	zip_code	total_shops	avg_stars	total_reviews	avg_reviews_per_shop	open_shops	closed_shops	closure_rate_pct
1	37220	3	4.67	93	31	2	1	33.3
2	37206	31	4.24	4125	133	19	12	38.7
3	37216	13	4.19	626	48	9	4	30.8
4	37208	19	4.16	3102	163	10	9	47.4
5	37209	28	3.77	2250	80	22	6	21.4
6	37204	22	3.77	3502	159	16	6	27.3
7	37201	15	3.77	3052	203	10	5	33.3
8	37219	21	3.76	1961	93	12	9	42.9

8 rows | 0.84s runtime

Finding	Hypothesis	Result	Implication for CoffeeKing
Downtown (37203) has most shops but lower ratings	Supported ✓	3.65 avg stars, 78 shops	Avoid downtown saturation
High-review shops score higher, not lower	Disproved ✗	500+ reviews = 4.12 stars	Quality drives both reviews and ratings
7-day shops score lower than 6-day shops	Disproved ✗	Mon-Sat = 4.26 vs 3.57	Open 6 focused days, not 7 mediocre ones
Best location: zip 37206	Data-driven ✓	4.24 stars, 31 shops, 4,125 reviews	High demand, proven quality market

Approach summary: SQL techniques applied include window functions (SUM() OVER()), CASE WHEN bucketing for continuous variable segmentation, CAST for data type correction, HAVING for aggregate-level filtering, LOWER() with LIKE for case-insensitive category matching, and multi-criteria ORDER BY for location ranking by quality and survival rate.

End of Milestone 1. Proposal subject to revision as analysis progresses